FeelPath Smart Cane - Components List (28 October 2026)

Prototype bill of materials | Badr Akoum

The prototype is a standalone smart cane: a downward LiDAR detects stairs, holes and curbs ahead, an upward ultrasonic sensor detects head-level obstacles, an IMU tracks the cane's angle, and two vibration motors in the handle alert the user. All processing runs on the cane's microcontroller.

1. Items in the Taobao cart

All electronics are in one shared cart: https://m.tb.cn/h.jUga7cS (Taobao code: Lz2YTBVcZPR). Cane: listing link (code: HRdvTBeEHJM). Links open in the Taobao app.

Item	Qty	Unit price	Purpose	Link
Benewake TF-Luna ToF LiDAR (UART)	1	¥130.00	Ground sensor aimed ~1.5 m ahead: detects stairs, holes, curbs and obstacles in the path.	Taobao cart
JSN-SR04T waterproof ultrasonic (board + 2.5 m probe)	1	¥16.10	Head-level obstacles, aimed ~45° up. Wide cone; also detects glass, which the LiDAR misses.	Taobao cart
MPU6050 IMU (GY-521, original chip)	1	¥14.86	Measures cane tilt so distance readings can be interpreted: drop-off vs. tilted cane, obstacle height, ignoring cane lifts.	Taobao cart
DRV2605L haptic driver	2	¥16.83	One driver per vibration motor, so each motor can play its own pattern.	Taobao cart
1040 LRA vibration motor	2	¥4.11	Vibration in the handle: the user's feedback channel (distinct patterns per hazard).	Taobao cart
MT3608 boost converter (2.5-15 V in)	2	¥4.14	Raises battery voltage (3.3-4.2 V) to a steady 5 V for the LiDAR and ultrasonic. One spare.	Taobao cart
LiPo 3.7 V 1000 mAh (4025100, with protection board)	1	¥5.28	Power supply, about 5 h runtime. Long thin shape fits inside the handle.	Taobao cart
Foldable white cane (reflective, telescopic)	1	¥19.30	Base for the prototype: sensors mount on the shaft, electronics in the handle.	Taobao listing

Cart total: about ¥236 (electronics about ¥216 + cane ¥19.30; prices after shop discounts, may change).

2. Items to source from the FabLab

Note: the following items are not in the cart. Their availability needs to be checked in the FabLab before ordering.

Item	Qty	Purpose
Seeed XIAO ESP32-S3 (plain, not the camera "Sense" version)	1	Microcontroller running all detection logic. Has two I2C buses, needed because both DRV2605L drivers share the same fixed address.
Slide switch	1	On/off switch in the battery's + wire.
Resistors: 1 kOhm, 2 kOhm, 2x 200 kOhm	1 set	1k + 2k: voltage divider for the ultrasonic's 5 V echo signal. 2x 200k: battery-level measurement for a low-battery vibration warning.
Electrolytic capacitor 220-470 µF, 16 V	1	Smooths the 5 V supply (the TF-Luna draws 150 mA peaks).
Breadboard + Dupont jumper wires (M-M, M-F, F-F)	1 set	Prototype wiring before soldering.
USB-C data cable	1	Programming and charging the XIAO.
Mounts and handle enclosure (3D printed)	-	Adjustable-angle brackets for the LiDAR and ultrasonic, a rain hood for the TF-Luna (not waterproof), and a handle case for electronics and battery.